

Comparative Analysis of Machine Learning Classifiers on the Iris Dataset

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Abstract—Machine learning classification algorithms are widely used for predictive analysis and pattern recognition. This paper presents a comparative analysis of multiple supervised machine learning classifiers on the Iris dataset using K-fold cross-validation. The classifiers considered include the Support Vector Machine (SVM), Random Forest, Decision Tree, Naive Bayes, and Multilayer Perceptron (MLP) with varying hidden layers. Performance evaluation is conducted using mean accuracy, standard deviation, R^2 score, confusion matrix analysis, and epoch-based learning curves. The experimental results demonstrate that SVM achieves the highest classification accuracy and stability among all models evaluated.

Index Terms—Machine Learning, Iris Dataset, SVM, Random Forest, Naive Bayes, Decision Tree, MLP, Cross Validation

I. INTRODUCTION

Classification is one of the most important tasks in machine learning and artificial intelligence. The Iris dataset is a widely used benchmark for evaluating classification algorithms due to its simplicity and balanced multiclass structure.

This research compares the performance of classical machine learning models and neural network-based classifiers using K-fold cross-validation. The objective is to identify the most accurate and stable classifier for the Iris dataset.

II. DATASET AND PREPROCESSING

The Iris dataset contains 150 samples with four numerical features and three classes of flower species:

- Setosa
- Versicolor
- Virginica

The features are:

- 1) Sepal Length
- 2) Sepal Width
- 3) Petal Length
- 4) Petal Width

Feature standardization was applied prior to training the SVM and MLP classifiers.

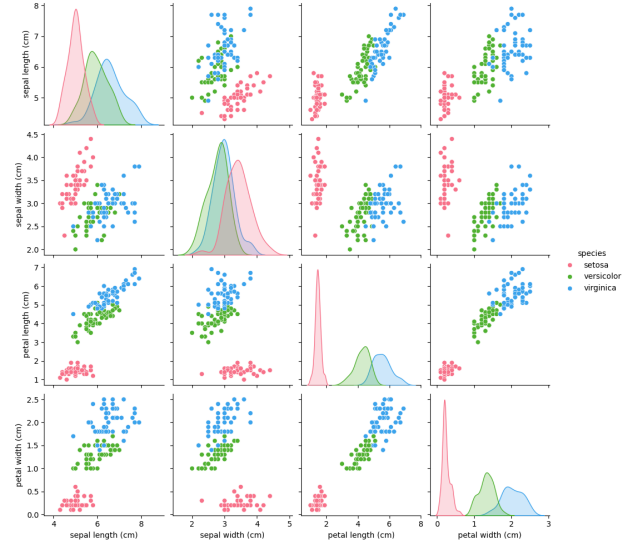


Fig. 1. Pairwise feature distribution of the Iris dataset

III. METHODOLOGY

The following classifiers were implemented:

- Support Vector Machine (Linear Kernel)
- Random Forest Classifier
- Decision Tree Classifier
- Gaussian Naive Bayes
- MLP with 1 Hidden Layer
- MLP with 2 Hidden Layers
- MLP with 3 Hidden Layers

K-Fold Cross Validation was performed for:

$$K = 2 \text{ to } 10$$

Performance metrics used:

- Mean Accuracy
- Standard Deviation
- R^2 Score
- Confusion Matrix
- Loss vs Epoch
- Accuracy vs Epoch

TABLE I
COMPARISON OF MEAN ACCURACY FOR DIFFERENT K-FOLDS

K-Fold	SVM	RF	DT	NB	MLP-1	MLP-2	MLP-3
2	0.9800	0.9600	0.9533	0.9533	0.8333	0.8667	0.7600
3	0.9733	0.9533	0.9533	0.9533	0.8133	0.8533	0.6867
4	0.9732	0.9532	0.9331	0.9467	0.8129	0.8396	0.7470
5	0.9667	0.9467	0.9533	0.9467	0.8200	0.8267	0.7400
6	0.9533	0.9400	0.9267	0.9467	0.8333	0.8267	0.7267
7	0.9527	0.9598	0.9329	0.9533	0.8126	0.8392	0.7393
8	0.9529	0.9463	0.9262	0.9529	0.8322	0.8392	0.7317
9	0.9530	0.9465	0.9261	0.9465	0.8325	0.8391	0.7398
10	0.9467	0.9467	0.9333	0.9533	0.8400	0.8400	0.7200

IV. EXPERIMENTAL RESULTS

A. Cross Validation Performance

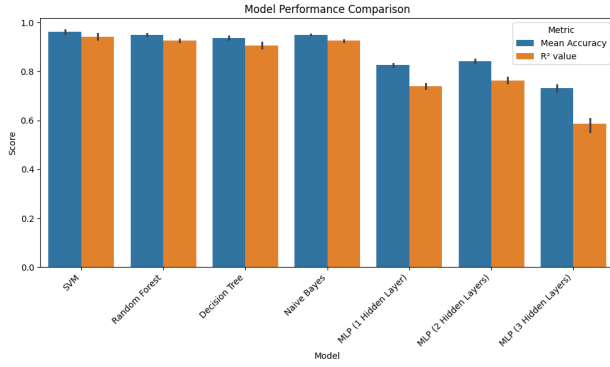


Fig. 2. Model Performance Comparison

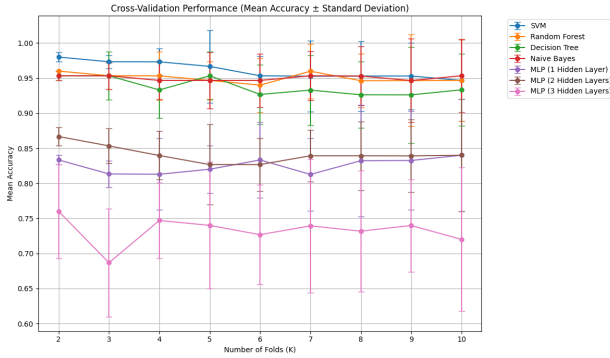


Fig. 3. Cross Validation Performance with Standard Deviation

B. Confusion Matrix Analysis

The confusion matrix analysis shows that SVM, Random Forest, and Naive Bayes correctly classify almost all test samples. MLP models exhibit comparatively lower performance due to increased model complexity.

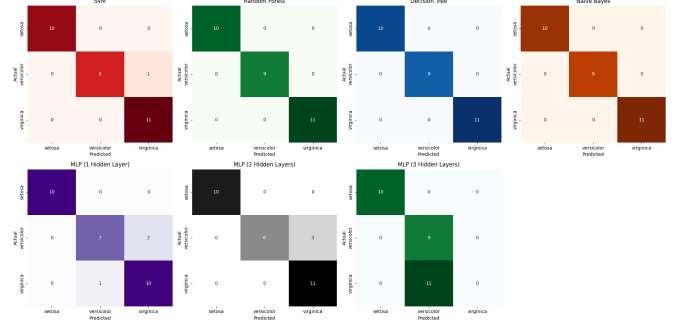


Fig. 4. Confusion Matrix Comparison of Different Models

C. MLP Loss and Accuracy Analysis

The training behavior of MLP models was analyzed using epoch-based learning curves.

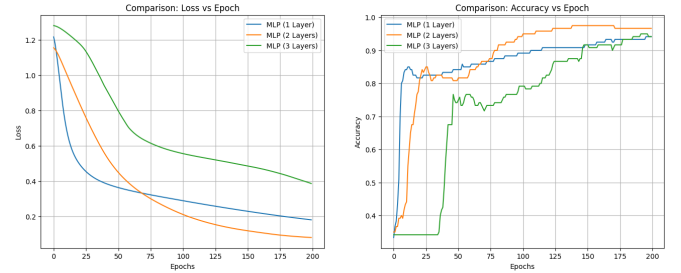


Fig. 5. Loss vs Epoch and Accuracy vs Epoch

V. DISCUSSION

Experimental results demonstrate that SVM achieved the best overall performance with a maximum mean accuracy of 98% and R^2 score of 0.97.

Random Forest and Naive Bayes also achieved consistently high accuracy across different K-Fold configurations. Decision Tree performance was slightly unstable because of its sensitivity to training data splits.

The MLP classifiers showed that increasing the number of hidden layers does not necessarily improve performance on small datasets. The 3-hidden-layer MLP showed higher variance and lower stability, indicating possible overfitting.

The descending order of classifier performance is:

SVM > Random Forest > Naive Bayes > Decision Tree > MLP (2 Layers) > MLP (1 Layer) > MLP (3 Layers)

VI. CONCLUSION

This study compared multiple supervised machine learning classifiers on the Iris dataset using K-fold cross-validation. The results show that SVM with a linear kernel provides the best combination of accuracy, robustness, and stability.

Future work may include:

- Hyperparameter optimization
- Deep learning architectures
- Ensemble stacking methods
- Evaluation on larger datasets

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